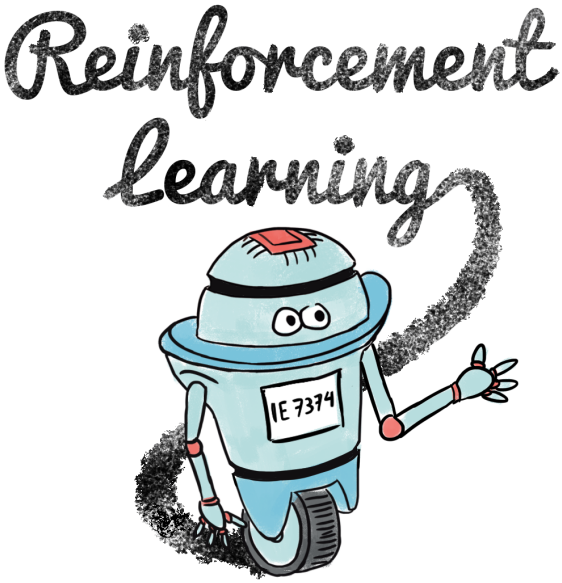
IE7295: Applied Reinforcement Learning



Prof. Mohammad Dehghani

"Sell your cleverness and buy bewilderment. Cleverness is mere opinion, bewilderment is intuition"
- Rumi

Northeastern University

Applied Reinforcement Learning

IE7295

Instructor Info

Mohammad Dehghani

Office Hrs: Thu 2:00pm-3:30pm

Zoom (make an appointment)

Website

m.dehghani@northeastern.edu

Course Info

Prerequisite:
* IE 6200 - Engineering Probability and Statistics.
* A previous exposure to the basic paradigms of algorithm design and analysis and familiarity with Python are essential.

Mon

6:00pm - 9:30pm

Behrakis Health Sciences Cntr Room 007

TA Info

Yash Nema

Office Hrs: Thu & Fri 6:00pm-8:00pm

Zoom (make an appointment)

nema.y@northeastern.edu

Course Description

Reinforcement Learning (RL) is a machine learning approach that aims to formulate an optimal policy from the trial-and-error learning process that is needed in many real-world applications in which no explicit instructive information is available. It has roots in operations research, behavioral psychology and artificial intelligence. The goal of the course is to introduce the basic mathematical foundations of reinforcement learning, as well as highlight some of the recent directions of research. The application of Reinforcement Learning in different domains including: manufacturing, robotics, healthcare, game playing, etc, will be discussed.

Course Objectives

This course is primarily designed to introduce reinforcement learning to graduate level students. Students will learn different methods that are used for learning based on feedback. On completion of the course, students should be able to:

- Identify basic concepts, terminology, theories, models and methods in reinforcement learning
- Develop and systematically test a number of basic methods in reinforcement learning
- Evaluate different learning algorithms experimentally and interpret and document results of experimental studies
- account for basic methods and limitations in reinforcement learning
- Apply reinforcement learning on real-life problems in order to be familiar with its basic possibilities and limitations and thereby be able to assess which problems can be solved with these technologies
- Be able to implement, analyze and evaluate simple systems based on reinforcement learning

Reference Books

Main Textbook

1. *Richard S. Sutton and Andrew G. Barto, Reinforcementt Learning: An Introduction' MIT Press, Second edition , 2018.* PDF.

Additional References (Optional)

1. Csaba Szepesvari, 'Algorithms for Reinforcement Learning', Morgan and Claypool, 2010. Link
2. Lucian Busoniu, Robert Babuska, Bart De Schutter, Damien Ernst *'Reinforcement Learning and Dynamic Programming Using Function Approximators'*, CRC Press, 2017. Link
3. Aurlien Gron, *'Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems'*, O'Reilly Media, First edition , 2017. Link

Course Outline

The following is a *tentative* schedule of the course.

- Introduction to Reinforcement learning
- Part 1: Tabular Solution Methods
  - Multi-armed Bandits (MABs)
  - Markov Decision Processes (MDPs)
  - Dynamic Programming (DP)
  - Monte Carlo Methods (MCM) and Bootstrapping
  - Reinforcement learning with Temporal Difference (TD), i.e.Sarsa and Q-learning
  - Planning and Learning with Tabular Methods (Dyna, Dyna-Q)
- Part 2: Prediction with Approximate Solution Methods
  - Value-function Approximation
  - Artificial Neural Networks (Deep Reinforcement Learning)
  - Semi-gradient n-step Sarsa
  - Control with Approximate Solution Methods

Software Tools

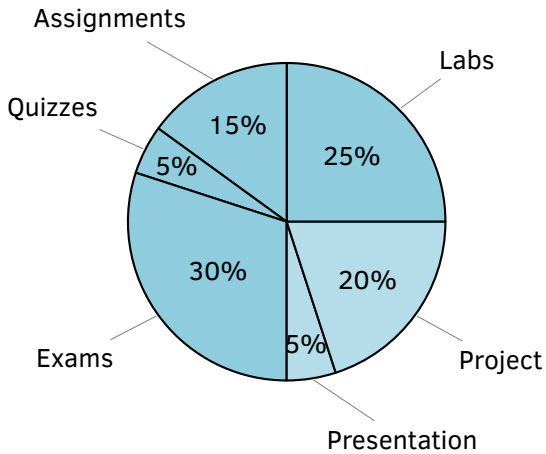
You need to have access to a python development environment for this course.

- Google Colab
  - The environment is already set up. There is no installation of packages required. You can run any python notebook on this environment. You need an internet connection to run the code. Link
- Jupyter-notebook-Local Development Environment
  - Please download a conda manager, either anaconda or miniconda. You can go ahead and download the environment.yml file and then run the following command on the terminal. In Windows 10, it is cmd or powershell.

```
$conda env create -f environment.yml
```

This created the conda environment with the name IE7295 with the base packages required for this course.

Grading Scheme



- Assignments (Individual)
  - Students will answer theoretical questions based on the coursework.
- Labs (Individual)
  - Students will be asked to customize or extend the algorithms explained in the labs for different needs.
- Quizzes (Individual)
  - Students will be required to complete these take-home quizzes on Canvas. Questions will be from the to be taught chapter(s).
- Exams: Midterm and Final Exam (Individual)
  - Both exams are equally weighted.
- Project (Individual or Group based)
  - Students will work in groups (2-3 memebrs) to apply RL algorithms to solve IE-related or real-life problems. Each group can include 2-3 members.Groups will present their project in the class in two-stages: *(1) Project proposal presentation*, *(2) Final Presentation*
  - The project timeline will be announced as the course progresses.
- Presentation (Group based)
  - Groups will be asked to have a short presentation related to the coursework in class. The topic for the presentation will be provided.

Grading

Your grades will be posted on CANVAS. You can check your running grade at any time, so there are no surprises. In fact, you should audit the correct entry of your grades. No grade corrections will be made after the last week of the semester. Final Grades and Course Letter Grades are not curved and are assigned as follows:

Letter Grade	Points	Letter Grade	Points	Letter Grade	Points
A	93-100	B	83-86	C	73-76
A-	90-92	B-	80-82	C-	70-72
B+	87-89	C+	77-79	F	<70

Emailing Us?

Please use the following format for the subject line of your email to avoid miss communication. This subject line will reduce the confusion.

IE7295 - Sec # - Subject Line

Make-up Policy

- No makeup opportunity will be given for any homeworks/lab assignment/projects.
- Students unable to attend classes are responsible for obtaining information regarding announcements and lecture content

Class Policy

- Students are expected to attend all class sessions and are responsible for all materials presented in class. Past experience has shown students who attend regularly and interact with instructors/classmates tend to be better overall.
- Laptops: You may need to use your laptop during the class upon the instructor's notes
- Phones/tablets/smart watches: Electronic devices may not be used during the class. If you have class work, social networking or sports-score browsing, that is more important to do, then please skip class. Life is full of such choices.
- Discussion between students for homeworks and projects is permissible. However the ideas / solutions / codes must not be copied. Each individual submission must be an outcome of individual effort

Recording of Classes

Classes will be recorded to enable all students to review material covered in synchronous classes. Please contact me if you have any concerns.

Academic Integrity

A commitment to the principles of academic integrity is essential to the mission of Northeastern University. The promotion of independent and original scholarship ensures that students derive the most from their educational experience and their pursuit of knowledge. Academic dishonesty violates the most fundamental values of an intellectual community and undermines the achievements of the entire University.

As members of the academic community, students must become familiar with their rights and responsibilities. In each course, they are responsible for knowing the requirements and restrictions regarding research and writing, examinations of whatever kind, collaborative work, the use of study aids, the appropriateness of assistance, and other issues. Students are responsible for learning the conventions of documentation and acknowledgment of sources in their fields. Northeastern University expects students to complete all examinations, tests, papers, creative projects, and assignments of any kind according to the highest ethical standards, as set forth either explicitly or implicitly in this Code or by the direction of instructors.

Go to Academic Integrity website to access the full academic integrity policy.

Student accommodations

Northeastern University and the Disability Resource Center (DRC) are committed to providing disability services that enable students who qualify under Section 504 of the Rehabilitation Act and the Americans with Disabilities Act Amendments Act (ADAAA) to participate fully in the activities of the university. To receive accommodations through the DRC, students must provide appropriate documentation that demonstrates a current substantially limiting disability.

For more information, visit Disability Resource Center website

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Diversity and Inclusivity Statement

Northeastern University is committed to equal opportunity, affirmative action, diversity and social justice while building a climate of inclusion on and beyond campus. In the classroom, members of the University community work to cultivate an inclusive environment that denounces discrimination through innovation, collaboration and an awareness of global

perspectives on social justice. It is my intention that students from all backgrounds and perspectives will be well served by this course, and that the diversity that students bring to this class will be viewed as an asset. I welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, socioeconomic background, family education level, ability - and other visible and nonvisible differences. All members of this class are expected to contribute to a respectful, welcoming and inclusive environment for every other member of the class. Your suggestions are encouraged and appreciated.

Please visit the website for complete information on Diversity and Inclusion.

TITLE IX

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance. Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.
In case of an emergency, please call 911.

Please visit the website for a complete list of reporting options and resources both on- and off-campus